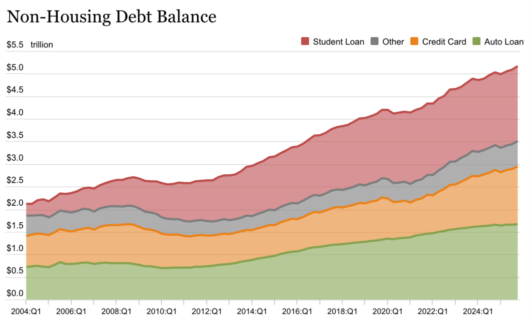
Self-Control and Commitment in Consumer Credit Markets

Ke Shi[1] and Yi Xin[2]

[1]Shanghai Advanced Institute of Finance (SAIF), Shanghai Jiao Tong University
[2]California Institute of Technology

Introduction

Background: Rising Household Debt



Source: Federal Reserve Bank of New York, Quarterly Report on Household Debt and Credit

- Household debt (excluding mortgages) in the US has grown steadily since 2000
- By late 2024, total non-housing household debt reached ≈ $5 trillion

Motivation: Debt Consolidation/Refinancing and the “Debt Trap”

Q1 2026 Unsecured Personal Loan Trends

Personal Loan Metric	Q1 2026	Q1 2025	Q1 2024	Q1 2023
Total Balances	**$277 billion**	$253 billion	$245 billion	$225 billion
Number of Unsecured Personal Loans	**32.6 million**	29.8 million	28.1 million	26.9 million
Number of Consumers with Unsecured Personal Loans	**26.4 million**	24.6 million	23.5 million	22.4 million

Source: TransUnion, Quarterly Credit Industry Insights Report (CIIR)

- Fintech boom in **debt consolidation/refinancing**: cheaper refinancing of high-APR balances; unsecured personal loans nearly doubled since 2020
- **Paradox:** P2P/fintech lending is associated with *more* bankruptcies (Wang and Overby 2022)

Self-Control as a Credit-Market Friction

- **Mechanism:** cash disbursement creates a temptation window
  - Ex-ante plan: consolidate debt and repay
  - Ex-post impulse: cash on hand $\rightarrow$ discretionary spending
  - Overspending raises the debt burden $\rightarrow$ "debt trap" $\rightarrow$ higher default
- **Distinctive friction:** an *internal* conflict (planner vs. doer)
  - Unlike adverse selection / moral hazard, the conflict is not lender vs. borrower
  - Both lender and borrower want to curb misuse
- **Implication:** because interests align, the lender can offer a **commitment device** that borrowers willingly adopt

Research Question and Setting

- **Questions**
  1. How costly are self-control problems in unsecured personal loans for debt consolidation/refinancing?
  2. How much can a commitment device improve repayment, net of selection and pricing?
- **Setting:** large U.S. fintech lender
  - Menu design change in 2018 around disbursement options
  - **Cash (status quo):** lump sum to borrower's bank; can divert to consumption
  - **Direct-pay (new):** funds wired to creditors; never touches cash (+ rate discount)
- Now an industry-standard feature among major fintech lenders

Preview of Findings

1. **Model:** present-biased refinancing with planner–doer conflict; direct-pay is jointly a *commitment* device and a *screening* device

2. **Reduced form**
   - **ITT:** offering the option cuts eligible defaults by 0.6 pp ($\approx$ 11%)
   - Effects concentrated among risky but liquid borrowers (low FICO, high income / low DTI)
   - **Adopters:** cash–direct gap of 3.6 pp; $\approx 3/4$ residual improvement vs. $\approx 1/4$ selection (illustrative decomposition)

3. **Structural**
   - Dynamic model of disbursement choice and repayment decisions
   - Self-control accounts for $\approx$ **35%** of pre-policy defaults
   - Of the cash–direct gap: selection 8%, commitment 53%, rate discount 38%
   - Counterfactuals: mandatory routing, upfront adoption bonuses

Related Literature

1. **Behavioral household finance / self-control** (Campbell 2006; Beshears et al. 2018; Gomes, Haliassos, and Ramadorai 2021; Meier and Sprenger 2010; DellaVigna and Malmendier 2004)
   - Field evidence on self-control in debt consolidation (cash vs. direct-pay); quantify how present-biased diversion contributes to default

2. **Commitment devices** (Bryan, Karlan, and Nelson 2010; Ashraf, Karlan, and Yin 2006; Kaur, Kremer, and Mullainathan 2015; Vihriälä 2023)
   - Lender-designed commitment at origination in debt refinancing; separate behavioral gains from sorting and pricing

3. **Screening under asymmetric information** (Einav, Jenkins, and Levin 2012; Kawai, Onishi, and Uetake 2022; Yannelis and Zhang 2023; Xin 2020)
   - Disbursement menu as both screening *and* commitment; jointly quantify selection and behavioral channels

Theoretical Framework

Model: Debt Refinancing with Temptation

- Outstanding debt s; borrower takes a refinancing loan of size s
- Timeline
  - $t = 0$: choose disbursement: cash vs. direct-pay
  - $t = 1$: fund arrives
    - **Abstain:** repay old debt in full; debt burden stays s
    - **Spend:** divert share α; enjoy $u(\alpha s)$; debt rises to $(1 + \alpha)s$
  - $t = 2, \ldots, T$
    - Repay installment m
    - Default with cost c (private type / creditworthiness)
- Preferences: quasi-hyperbolic (β, δ)

$$U_t = u(x_t) + \beta \sum_{\tau > t} \delta^{\tau - t} u(x_\tau), \qquad \beta \in (0, 1]$$

Preference Reversal: Planner vs. Doer

- Let $V_1(\cdot)$ be the continuation value from the repayment subgame
- Ex-ante $(t = 0)$ **planner**:

$$U_0^{\text{spend}} = \beta\big(\delta u(\alpha s) + \delta^2 V_1^{\text{spend}}\big), \quad U_0^{\text{abstain}} = \beta\delta^2 V_1^{\text{abstain}}$$

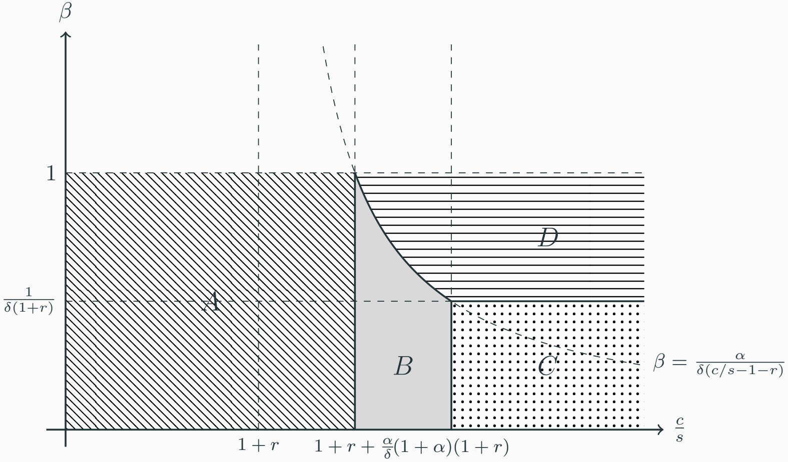
- Ex-post $(t = 1)$ **doer**:

$$U_1^{\text{spend}} = u(\alpha s) + \beta\delta V_1^{\text{spend}}, \quad U_1^{\text{abstain}} = \beta\delta V_1^{\text{abstain}}$$

- **Self-control problem** $=$ preference reversal:

$$U_0^{\text{abstain}} > U_0^{\text{spend}} \quad \text{but} \quad U_1^{\text{spend}} > U_1^{\text{abstain}}$$

Who Has a Self-Control Problem?



Direct-Pay: Commitment *and* Screening

- **Commitment:** sophisticated planner at $t = 0$ elects direct-pay $\Rightarrow$ forces $\alpha = 0$ at $t = 1$
- **Prediction 1:** offering the option should lower *aggregate* defaults via behavior
- **Screening:** c is private information
  - High-c types value commitment (and repayment) more $\Rightarrow$ sort into direct-pay
  - Low-c types prefer cash flexibility $\Rightarrow$ remaining cash pool worsens
  - Separating equilibrium on default cost (single-crossing in V_1)
- **Prediction 2:** relative to 2017 (cash-only), 2018 cash defaults rise; direct-pay defaults fall
  - Observed cash–direct gap mixes **selection** and **behavior** (+ pricing)

Institutional Background & Data

Institution and Data

- Unsecured personal loans up to \$40k; terms 3 or 5 years; mean rate $\approx$ 13–14%
- $\approx$ 80% debt-refinancing / credit-card repayment; remainder general-purpose consumption
- Rich underwriting covariates: income, DTI, FICO, credit history, revolving utilization, etc.
- Outcome: one-year default (sample mean $\approx$ 5.7%)
- Direct-pay introduced in 2018 for refinancing loans only
  - Take-up $\approx$ 1/4 of eligible borrowers
  - Pilot-era interest discount (recovered $\approx$ 3.5 pp on average); not randomly assigned
  - We treat pricing as a feature of the rollout and separate it structurally

Empirical Evidence

ITT: Effect of Policy *Availability*

- DiD on the same platform:
  - Treated: refinancing loans (eligible in 2018)
  - Control: consumption loans (never eligible)

$$Default_i = \beta_1 Post_t + \beta_2 Treated_i + \beta_3(Post_t \times Treated_i) + X_i'\gamma + \mu_t + \varepsilon_i$$

- β_3: ITT of *eligibility*, not adoption
- **Result:** -0.6 pp $\approx$ **11%** decline relative to pre-policy mean ($\approx$ 5.2%)
- Identification: parallel pre-trends (event study 2014–17); stable observables across eligibility and years

Heterogeneity: “Risky but Liquid”

- Strongest DiD effects among borrowers who look
  - **Risky:** low FICO ($\approx -1.0\,$pp)
  - **Liquid:** high income; low DTI
  - Medium revolving utilization (scope to add discretionary debt)
- Interpretation: gains concentrate where means to repay coexist with discipline failures, not pure liquidity constraints

Adopters: Selection vs. Behavior

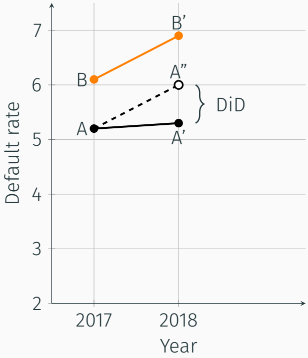
- Among 2018 eligible loans: cash default 6.2% vs. direct-pay 2.6% $\Rightarrow$ raw gap **3.6** pp
- Challenge: gap $=$ selection $+$ commitment $+$ rate discount
- Use 2017 eligible (cash-only) cohort as product-specific baseline

$$Default_i = \beta_1 PostCash_i + \beta_2 PostDirect_i + X_i'\gamma + \mu_t + \varepsilon_i$$

  - $PostCash > 0$: remaining cash pool worsens (adverse selection), as theory predicts
  - $PostDirect < 0$: adopters improve by more than sorting alone can explain
- Next: visualize the decomposition with the 2017 type-specific counterfactuals

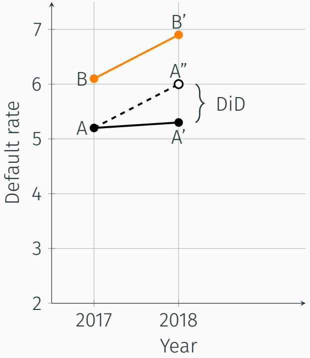
Default Dynamics: ITT and Decomposition

(a) Policy Effect
Eligible vs **Ineligible**

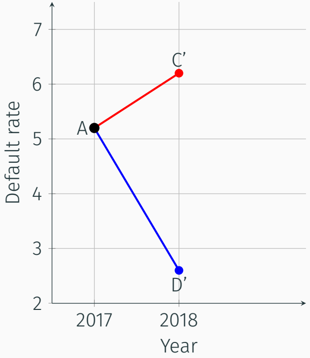


Default Dynamics: ITT and Decomposition

(a) Policy Effect
Eligible vs **Ineligible**

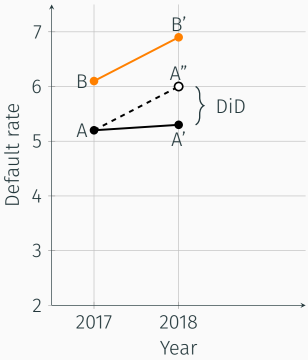


(b) Decomposition
Cash vs **Direct-Pay** (eligible)

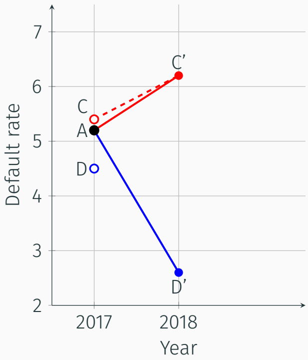


Default Dynamics: ITT and Decomposition

(a) Policy Effect
Eligible vs **Ineligible**

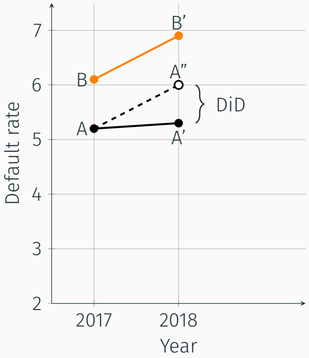


(b) Decomposition
Cash vs **Direct-Pay** (eligible)

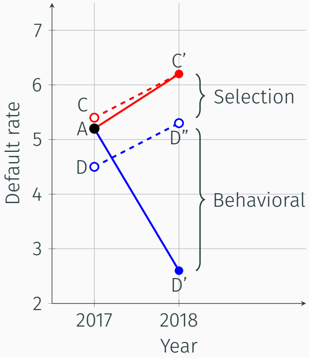


Default Dynamics: ITT and Decomposition

(a) Policy Effect
Eligible vs **Ineligible**



(b) Decomposition
Cash vs **Direct-Pay** (eligible)



Structural Model & Estimation

Why a Structural Model?

1. Reduced-form residual mixes **commitment** and the **interest-rate discount**; the discount also shifts take-up and pool composition
2. Need a joint model of menu choice and repayment to separate selection, commitment, and pricing
3. Enables counterfactuals beyond the observed policy
   - How large is the self-control friction in defaults?
   - Alternative contracts: mandatory routing, adoption bonuses, ...

Structural Model: Outline

- Binary private type $c \in \{c_L, c_H\}$; $P(H \mid X)$ logistic in observables
- Spending propensity $\alpha(X)$: proxy for self-control (higher $\alpha \Leftrightarrow$ worse present bias)
- $t = 0$ (origination): choose cash vs. direct-pay

$$\begin{cases} U^{\text{Cash}} = u(\alpha s + \omega) + V_1\big((1+\alpha)s,\ r_c,\ c\big) + \eta_0 \\ U^{\text{Direct}} = V_1(s,\ r_d,\ c) + \eta_1 \end{cases}$$

  - ω: non-pecuniary cash preference / adoption friction

Structural Model: Outline (Cont.)

- $t = 1, \ldots, T$: repay installment m or default (follows Kawai, Onishi, and Uetake 2022)
- Flow utilities ($t < T$; linear $u(x) = x$):

$$\begin{cases} u(-m) + \delta V_{t+1}(s, r, c) + \epsilon_{t,0} & \text{repay} \\ u(-c) + \epsilon_{t,1} & \text{default} \end{cases}$$

  - m is the monthly installment determined by (s, r, T)

Estimation and Identification

- Sample: 3-year loans in 2017–2018; GMM
- Four moments, each interacted with covariates X
  1. 2017 (pre-policy) default rate
  2. 2018 direct-pay take-up
  3. 2018 cash default rate
  4. 2018 direct-pay default rate
- What pins down what
  - $\{c_L, c_H\}$: default *levels*
  - $P(H \mid X)$, ω: take-up across X; composition of cash vs. direct-pay pools
  - $\alpha(X)$: residual cash–direct default gap, given selection and observed rates

Estimates

- Default costs: $c_L = \$2.7$k, $c_H = \$4.5$k
- Share high-type: $P(H) = 0.70$
  - higher income, FICO, longer history $\Rightarrow$ higher $P(H)$
- Mean discretionary share: $\alpha = 0.33$
- Cash preference: $\omega_L = \$0.39$k, $\omega_H = \$0.77$k
  - High types still sort into direct-pay: commitment value outweighs larger ω_H

How Large Are Self-Control Problems?

Scenario	Take-up	Default rate		
		Cash	Direct	Overall
(1) Rational benchmark ($\alpha = 0$)	–	3.75%	–	3.75%
(2) Behavioral baseline (no intervention)	–	5.75%	–	5.75%
(3) Current policy	26.0%	5.82%	2.64%	5.00%

- Self-control gap: $5.75 - 3.75 = 2.00$ pp $\Rightarrow \approx$ **35%** of pre-policy defaults
- Current policy: aggregate default $5.75\% \to 5.00\%$

Decomposition: Selection, Commitment, Pricing

Scenario	Take-up	Default rate		
		Cash	Direct	Overall
(2) Behavioral baseline	–	5.75%	–	5.75%
(4) Selection only (no improvement)	26.0%	5.82%	**5.55%**	5.75%
(5) Direct-pay with zero discount	26.0%	5.82%	**3.85%**	5.31%
(3) Current policy	26.0%	**5.82%**	**2.64%**	5.00%

- Cash–direct gap under current policy: 3.18 pp
  - Selection 0.27 pp (**8%**); commitment 1.70 pp (**53%**); discount 1.21 pp (**38%**)
- Aggregate: of the 0.75 pp overall decline, commitment 0.44 pp and discount 0.31 pp (holding sorting fixed)
- Ranking matches reduced form: improvement $\gg$ selection

Alternative Contract Designs

Scenario	Take-up	Default rate		
		Cash	Direct	Overall
(3) Current policy	**26.0%**	5.82%	2.64%	**5.00%**
(6) Zero discount (endogenous take-up)	**23.3%**	5.87%	3.72%	5.37%
(7) Mandatory direct-pay	100%	–	3.01%	**3.01%**
(8) Upfront bonus ($\omega \downarrow \$200$)	**34.3%**	5.84%	2.68%	4.76%

- Discount is a weak adoption lever (take-up falls only to 23%); commitment still helps
- Upfront bonus raises take-up more than rate cuts (present-biased agents overweigh immediate rewards)
- Mandatory routing lowers defaults most, but removes liquidity; welfare not quantified

Conclusion

Conclusion

1. Self-control is first-order: $\approx$ **35%** of pre-policy defaults
2. Direct-pay improves repayment
   - Availability: -0.6 pp ($\approx$ 11%; ITT), concentrated among risky-but-liquid borrowers
   - Adopters improve far more than selection alone predicts
3. Mechanism: commitment dominates selection; interest discounts matter but are secondary
4. **Policy:** disbursement design is a practical lever. Restrict ex-post discretion over proceeds to curb misuse, raise repayment, and induce sorting on creditworthiness